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PROMETHEE Analysis

Preference Information Used as Input

The PROMETHEE method utilizes the following preference information to calculate marginal and comprehensive preference indices:

- **Criteria and Types:**

- **Cost:**
 - price,
 - mileage,
 - fuel_consumption
- **Gain:**
 - year,
 - power,
 - appearance

- **Weights (w):**

- price (5),
- fuel_consumption (4),
- mileage (4),
- power (3),
- appearance (3),
- and year (2).

- **Indifference (q) and Preference (p) Thresholds:**

- **Price:** q = 500, p = 2000
- **Year:** q = 0, p = 1
- **Mileage:** q = 5000, p = 20000
- **Power:** q = 10, p = 30
- **Fuel Consumption:** q = 0.3, p = 1.0
- **Appearance:** q = 1.0, p = 2.0

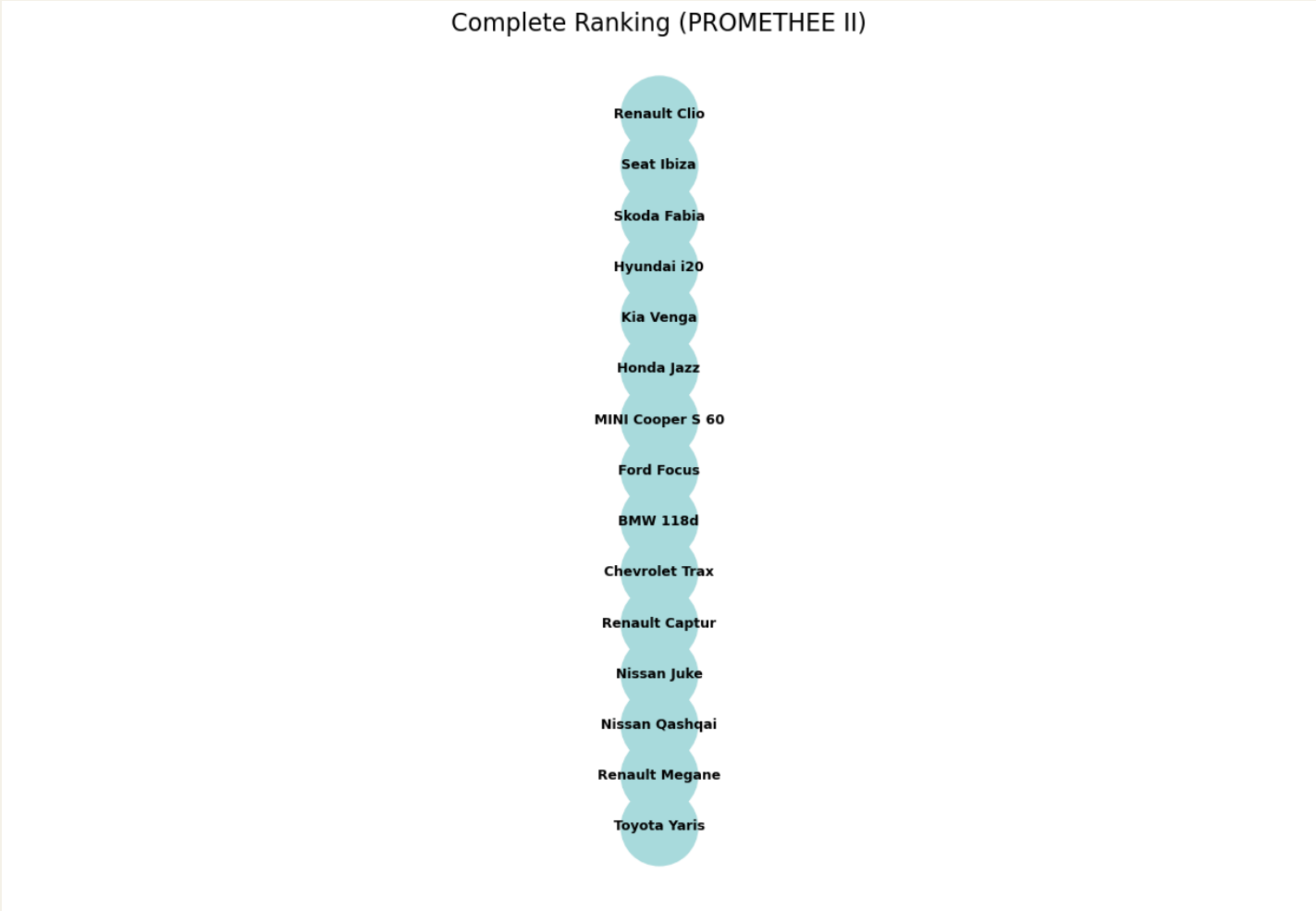
Final Results Obtained

The final PROMETHEE II ranking is determined by the **net flow (Φ)**, which is the difference between the positive flow (how strongly an alternative outranks others) and the negative flow (how strongly it is outranked).

Rank	Alternative	Net Flow (ϕ)
1	Renault Clio	0.334945
2	Seat Ibiza	0.170878
3	Skoda Fabia	0.167775
4	Hyundai i20	0.143591
5	Kia Venga	0.118918
6	Honda Jazz	0.091207
7	MINI Cooper S 60	0.051343
8	Ford Focus	-0.057905
9	BMW 118d	-0.060972
10	Chevrolet Trax	-0.085525
11	Renault Captur	-0.094093
12	Nissan Juke	-0.152211
13	Nissan Qashqai	-0.165800
14	Renault Megane	-0.188346
15	Toyota Yaris	-0.273805

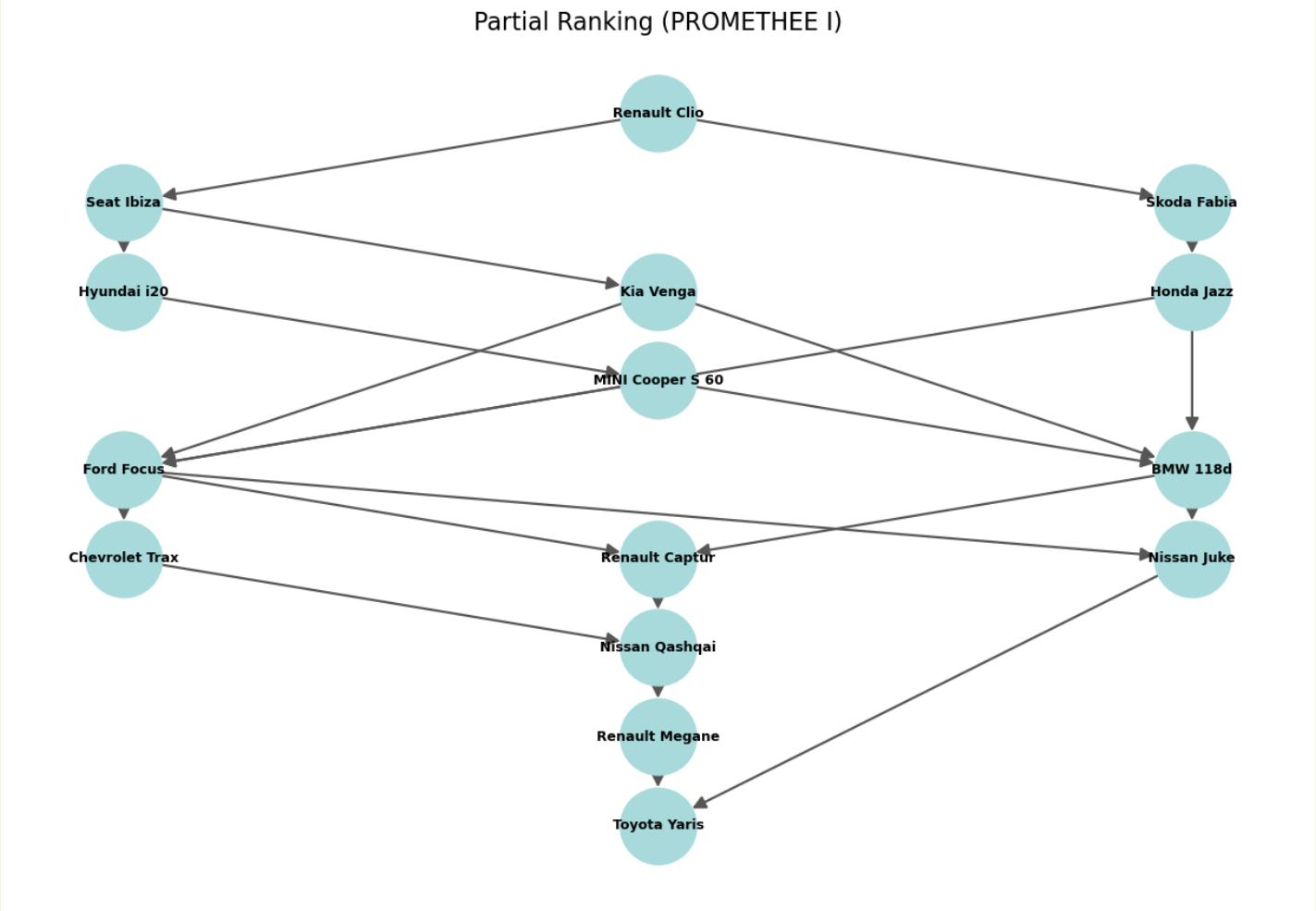
Comparison of Complete and Partial Rankings

- Complete Ranking (PROMETHEE II):** This relies purely on the net flow (ϕ) to enforce a strict transitive preorder, allowing every alternative to be directly compared to another.



- Partial Ranking (PROMETHEE I):** This ranking is generated by evaluating the positive flow (ϕ^+) and negative flow (ϕ^-) separately. If one alternative has a higher positive flow but a lower negative flow than another, they are deemed incomparable. The partial ranking preserves these incomparabilities, acknowledging that an alternative might be superior

in some major criteria but severely worse in others.



Findings and Pairwise Comparisons

- **Best and Worst Alternatives:** The best alternative is the **Renault Clio** (highest net flow at 0.334945), and the worst is the **Toyota Yaris** (lowest net flow at -0.273805).
- **Validation of Expected Pairwise Comparisons (from Project 1):**
 - *Hyundai i20 > Kia Venga:* **Satisfied.** Hyundai i20 (0.143) > Kia Venga (0.118).
 - *Renault Clio > Skoda Fabia:* **Satisfied.** Renault Clio (0.334) > Skoda Fabia (0.167).
 - *Hyundai i20 > Toyota Yaris:* **Satisfied.** Hyundai i20 (0.143) > Toyota Yaris (-0.273).
 - *Seat Ibiza > Ford Focus:* **Satisfied.** Seat Ibiza (0.170) > Ford Focus (-0.057).
 - *Kia Venga > Chevrolet Trax:* **Satisfied.** Kia Venga (0.118) > Chevrolet Trax (-0.085).
 - *Cycle (BMW 118d > Renault Megane > Nissan Qashqai > BMW 118d):* **Broken.** Because PROMETHEE calculates global net flows to build a strict hierarchy, it cannot accommodate cyclical preferences. The resulting order is BMW 118d (-0.060) > Nissan Qashqai (-0.165) > Renault Megane (-0.188).

ELECTRE Tri-B Analysis

Preference Information Used as Input

ELECTRE Tri-B utilizes the same base criteria, directions, weights, q, and p thresholds as PROMETHEE, but it introduces boundaries and veto thresholds for classification:

- **Boundary Profiles:**
 - **b1** (Separating Class 1 and Class 2): price: 37000, year: 2015, mileage: 120000, power: 110, fuel_consumption: 6.8, appearance: 6.

- **b2** (Separating Class 2 and Class 3): price: 33500, year: 2016, mileage: 80000, power: 120, fuel_consumption: 5.8, appearance: 7.
- **Veto Thresholds (v)**: Applied to cost criteria. Price (v=8000), mileage (v=80000), and fuel_consumption (v=3.0).
- **Credibility Threshold (λ)**: Set to **0.65** to validate outranking relations.

Final Results Obtained

The cars were sorted into three distinct classes: **C3 (Good)**, **C2 (Average)**, and **C1 (Weak)**.

Assignment	C3 (Good)	C2 (Average)	C1 (Weak)
Pessimistic	Renault Clio, Skoda Fabia	Kia Venga, Hyundai i20, Renault Megane, Chevrolet Trax, Toyota Yaris, Seat Ibiza, Renault Captur, Ford Focus, Honda Jazz	BMW 118d, Nissan Qashqai, MINI Cooper S 60, Nissan Juke
Optimistic	Renault Clio, Skoda Fabia	Kia Venga, Hyundai i20, BMW 118d, Renault Megane, Chevrolet Trax, Toyota Yaris, Seat Ibiza, Skoda Fabia, Renault Captur, Ford Focus, Honda Jazz	Nissan Qashqai, Nissan Juke

Comparison of Optimistic and Pessimistic Class Assignments

The two rules agreed on 13 out of 15 alternatives. Disagreements highlight alternatives that exhibit incomparability with the boundary profiles:

- **BMW 118d**: Assigned to C1 (Pessimistic) but C2 (Optimistic).
- **MINI Cooper S 60**: Assigned to C1 (Pessimistic) but C2 (Optimistic).

Findings and Pairwise Comparisons

- **Best and Worst**: The **Renault Clio** and **Skoda Fabia** are undeniably the best, landing in C3 under both assignment rules. **Nissan Qashqai** and **Nissan Juke** are the worst - in C1 under both assignment rules.
- **Validation of Expected Pairwise Comparisons**: Because ELECTRE Tri-B is a sorting method and not a ranking method, it does not strictly rank A over B; instead, it bins them.
 - *Renault Clio > Skoda Fabia*: **Tied** (C3).
 - *Hyundai i20 vs Kia Venga, Hyundai i20 vs Toyota Yaris, Seat Ibiza vs Ford Focus, Kia Venga vs Chevrolet Trax*: **Tied**. All these alternatives were placed into the C2 under both assignments. They are considered functionally equivalent in quality class relative to the defined boundary profiles.
 - *Cycle (BMW > Megane > Qashqai > BMW)*: Megane resolves to C2 reliably, while Qashqai stays at C1. BMW ranges from C1 to C2.

Method Comparison

Similarities and Differences

- **Goal Differences**: PROMETHEE provides a relative ranking (comparing alternatives directly against one another), which is excellent for finding a single "winner". ELECTRE Tri-B provides absolute sorting (comparing alternatives against predefined structural boundaries), making it ideal for tiering or filtering candidates.
- **Incomparability Handling**: PROMETHEE II mathematically aggregates differences into a single net flow, effectively masking incomparability to force a result. ELECTRE Tri-B embraces incomparability through the use of veto thresholds (preventing assignment to a high class if performance on a single criterion is disastrous) and dual-assignment rules (pessimistic vs. optimistic), giving the decision-maker deeper insight into controversial alternatives like the MINI Cooper S 60.
- **Threshold Mechanics**: Both methods rely on indifference (q) and preference (p) thresholds to build marginal outranking/preference indices, smoothing out minor data variances.